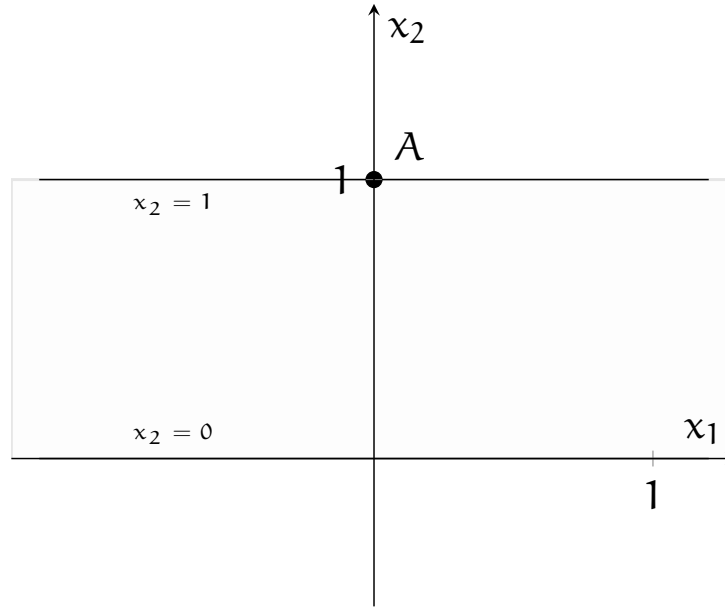


The constraint polyhedron is represented in the figure below:



In order to write the problem in standard form, we need to replace x_1 by the difference of two non negative variables

$$x_1 = x_1^+ - x_1^-,$$

and we need to introduce a slack variable for the constraint $x_2 \leq 1$. We obtain

$$\min_{x \in \mathbb{R}^4} x_1^+ - x_1^- + x_2$$

subject to

$$\begin{aligned} x_2 + x_3 &= 1, \\ x_1^+, x_1^-, x_2, x_3 &\geq 0. \end{aligned}$$

In matrix form, we have

$$\min_{x \in \mathbb{R}^4} c^T x$$

subject to

$$\begin{aligned} Ax &= b, \\ x &\geq 0, \end{aligned}$$

where

$$\begin{aligned} A &= \begin{pmatrix} 0 & 0 & 1 & 1 \end{pmatrix}, \\ b &= 1, \end{aligned}$$

$$c = \begin{pmatrix} 1 \\ -1 \\ 1 \\ 0 \end{pmatrix},$$

$$x = \begin{pmatrix} x_1^+ \\ x_1^- \\ x_2 \\ x_3 \end{pmatrix}.$$

This is a problem with $n = 4$ variables and $m = 1$ constraints. So the basis matrix is of dimension one, and is a scalar. If x_2 is in the basis, $B = 1$ and we have

$$x_2 = B^{-1}b = 1,$$

and the feasible basic solution is

$$\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix},$$

and corresponds to point A on the figure. As there are three non basic variables, there are three basic directions.

1. The basic direction associated with x_1^+ is

$$d_1^+ = \begin{pmatrix} 1 \\ 0 \\ d_B \\ 0 \end{pmatrix},$$

where

$$d_B = -B^{-1}A_1 = 0,$$

as the variable x_1^+ is associated with column 1 of A , so that

$$d_1^+ = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

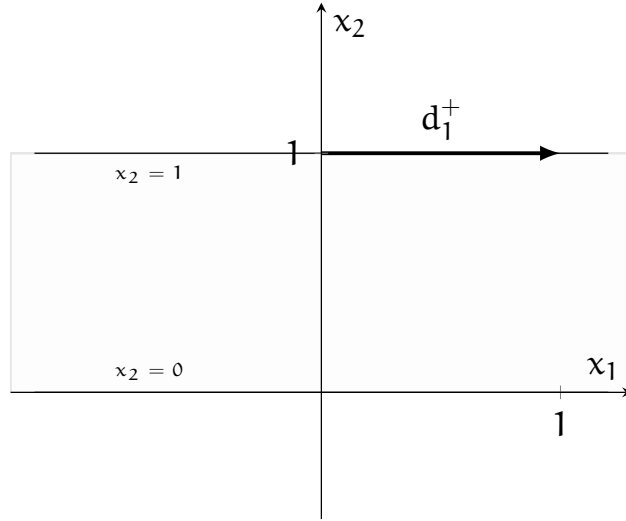
Performing a step α in this direction leads to the point

$$\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \alpha \\ 0 \\ 1 \\ 0 \end{pmatrix}.$$

The point is feasible for any value of $\alpha \geq 0$, so that the direction is feasible. In the original variables, with $x_1 = x_1^+ - x_1^-$, it corresponds to a direction

$$\begin{pmatrix} 1 - 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

as illustrated on the following figure:



2. The basic direction associated with x_1^- is

$$d_1^- = \begin{pmatrix} 0 \\ 1 \\ d_B \\ 0 \end{pmatrix},$$

where

$$d_B = -B^{-1}A_2 = 0,$$

as the variable x_1^- is associated with column 2 of A , so that

$$d_1^- = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

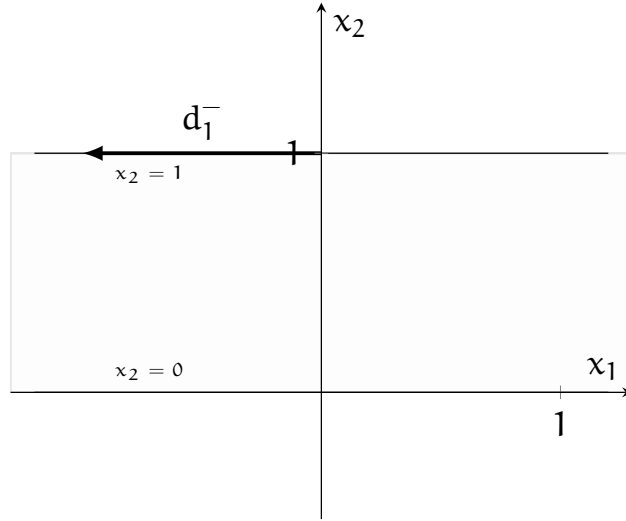
Performing a step α in this direction leads to the point

$$\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ \alpha \\ 1 \\ 0 \end{pmatrix}.$$

The point is feasible for any value of $\alpha \geq 0$, so that the direction is feasible. In the original variables, with $x_1 = x_1^+ - x_1^-$, it corresponds to a direction

$$\begin{pmatrix} 0 & -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix},$$

as illustrated on the following figure:



3. The basic direction associated with x_3 is

$$d_3 = \begin{pmatrix} 0 \\ 0 \\ d_B \\ 1 \end{pmatrix},$$

where

$$d_B = -B^{-1}A_4 = -1,$$

as the variable x_3 is associated with column 4 of A , so that

$$d_3 = \begin{pmatrix} 0 \\ 0 \\ -1 \\ 1 \end{pmatrix}.$$

Performing a step α in this direction leads to the point

$$\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} 0 \\ 0 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 - \alpha \\ \alpha \end{pmatrix}.$$

The point is feasible for any value of α such that $0 \leq \alpha \leq 1$, so that the direction is feasible. In the original variables, with $x_1 = x_1^+ - x_1^-$, it corresponds to a direction

$$\begin{pmatrix} 0 & -0 \\ -1 & \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix},$$

as illustrated on the following figure:

